

Step	Step Description	Agents Involved	Tool Calls and Parameters	Reason
1	User requests natural areas in Exeter	human, host_agent	Message sent: 'Show me natural areas in Exeter'	User initiates the query; host_agent receives and relays the request.
2	Plan steps to solve the query	planning_agent	N/A	Planning agent decomposes the query into steps: find Exeter area, search for natural areas within it, analyze and visualize results.
3	Generate metadata for all artifacts	host_agent	generate_metadata_for_all_artifacts {}	Ensures up-to-date knowledge of available artifacts for subsequent steps.
4	Find named area polygon for Exeter	host_agent, named_area_agent	send_message ({'target': 'named_area', 'task_description': 'Find the named area polygon for Exeter'})	named_area_agent is chosen because the prompt specifies it handles named area polygons; 'Exeter' is the area of interest.
5	Search named area polygons	named_area_agent	call_os_ngd ({'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_named_area'})	named_area was chosen because the prompt indicates it contains polygons of areas; no filters as we want all possible named areas for Exeter.
6	Filter for the specific Exeter polygon	named_area_agent, data_analysis_agent	send_message ({'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_named_area' to find the specific named area polygon for Exeter. Save the filtered results as 'filtered_exeter_named_area'."})	data_analysis_agent is chosen to filter the artifact; 'exeter_named_area' is used as input based on previous step.
7	Generate metadata for 'exeter_named_area'	data_analysis_agent	generate_metadata_for_artifacts ({'artifact_names': ['exeter_named_area']})	Metadata is needed to understand the columns and structure for correct filtering.
8	Filter code execution for Exeter polygon	coding_agent (via data_analysis_agent)	code_executor with code to filter for rows where any name column matches 'Exeter' and description is a settlement type; artifact_names: ['exeter_named_area']	Code uses name columns and settlement descriptions based on metadata and prompt requirements; ensures only the correct Exeter polygon is selected.
9	Confirm filtered Exeter polygon	data_analysis_agent	Output: 1 entry found, artifact 'filtered_exeter_named_area' generated	Confirms that the correct polygon for Exeter has been isolated.
10	Find natural land features in Exeter	host_agent, land_features_agent	send_message ({'target': 'land_features', 'task_description': "Find natural land features (such as coniferous trees, cliffs, heath, rough grassland, etc.) within the area defined by the artifact 'filtered_exeter_named_area'."})	land_features_agent is chosen because the prompt specifies it handles natural land features; bbox is set to 'filtered_exeter_named_area' to restrict search to Exeter.
11	Search for natural land features	land_features_agent	call_os_ngd ({'filters': ['Coniferous Trees', 'Cliff', 'Heath', 'Rough Grassland'], 'bbox': 'filtered_exeter_named_area', 'polygon_or_point': True, 'street_address': False, 'filename': 'natural_land_features_exeter'})	Filters chosen based on user request and available filter list in the agent prompt; bbox ensures features are within Exeter.
12	Analyze and combine land feature results	land_features_agent, data_analysis_agent	send_message ({'target': 'data_analysis_agent', 'task_description': "Please analyze the artifacts 'land_natural_land_features_exeter' and 'landform_natural_land_features_exeter' to provide insights on the natural land features such as coniferous trees, cliffs, heath, and rough grassland within the area defined by 'filtered_exeter_named_area'. Save the filtered results as 'analyzed_natural_land_features_exeter'."})	data_analysis_agent is chosen to combine and analyze results from both land and landform datasets; artifacts are named based on previous tool outputs.
13	Generate metadata for analysis	data_analysis_agent	generate_metadata_for_artifacts ({'artifact_names': ['land_natural_land_features_exeter', 'landform_natural_land_features_exeter', 'filtered_exeter_named_area']})	Metadata is needed to understand columns and ensure correct filtering and spatial operations.
14	Execute analysis code to filter and combine features	coding_agent (via data_analysis_agent)	code_executor with code to filter for relevant features (coniferous trees, heath, rough grassland, cliffs) and spatially restrict to Exeter polygon; artifact_names: ['land_natural_land_features_exeter', 'landform_natural_land_features_exeter', 'filtered_exeter_named_area']	Code uses description and oslandcovertierb columns for filtering, and spatial intersection with Exeter polygon; columns and logic chosen based on metadata and prompt.
15	Confirm analysis and artifact creation	data_analysis_agent	Output: 18 from land, 19 from landform, total 37 features; artifact 'analyzed_natural_land_features_exeter' generated	Confirms successful analysis and combination of features; artifact is ready for visualization.
16	Plot natural land features with Exeter boundary	host_agent, plotting_agent	send_message ({'target': 'plotting_agent', 'task_description': "Plot the natural land features in Exeter using the artifact 'analyzed_natural_land_features_exeter'. Add a boundary for the Exeter area using 'filtered_exeter_named_area'."})	plotting_agent is chosen for visualization; both analyzed features and boundary artifact are provided for context.
17	Generate metadata for plotting	plotting_agent	generate_metadata_for_artifacts ({'artifact_names': ['analyzed_natural_land_features_exeter', 'filtered_exeter_named_area']})	Metadata ensures correct columns and CRS for plotting.
18	Execute plotting code	coding_agent (via plotting_agent)	code_executor with code to plot features and boundary on a folium map, save as HTML; artifact_names: ['analyzed_natural_land_features_exeter', 'filtered_exeter_named_area']	Code uses folium for mapping, ensures CRS is EPSG:4326, and includes both features and boundary; columns chosen based on metadata.
19	Confirm plot and artifact creation	plotting_agent	Output: 37 features plotted, artifact 'natural_land_features_exeter_map' generated (HTML map)	Confirms successful visualization and artifact creation.
20	Final user output	host_agent	Output: Map and summary provided to user	Summarizes the results and offers the map for download or further queries.

Overall Summary: - Databases used: Named area (for Exeter boundary), land features and landform features (for natural areas). - Filters: For land features, filters included 'Coniferous Trees', 'Cliff', 'Heath', 'Rough Grassland'. - Code: Used to filter named area polygons for Exeter, spatially filter and combine natural features, and plot results on a folium map. - Final outcome: 37 natural land features (coniferous trees, cliffs, heath, rough grassland) within Exeter were found, analyzed, and visualized on an interactive map with the city boundary. - Reasoning: Each agent and tool was chosen based on its prompt-defined capabilities and the requirements of the user query. Parameters and columns were selected based on metadata and system prompts to ensure accurate filtering, analysis, and visualization.